

Walter, 72 · 15 minutes of right arm weakness and slurred speech that fully resolved · **left carotid bruit** on exam · carotid duplex: 80% stenosis · **TIA from carotid atherosclerosis**

BIO 004 · WEEK 3 · CASE DEEP DIVE

Blood Vessels, Structure & Types

A retired teacher who briefly couldn't speak. A bruit over the carotid. An artery wall that has been silently changing for decades.

PATIENT CASE

Walter is 72, retired English teacher, hypertensive, hyperlipidemic, 40-pack-year smoking history (quit a decade ago). This morning he had a sudden **15-minute episode of right arm weakness and slurred speech** that **completely resolved**. He came in once his wife insisted.

On exam he is at his neurological baseline. The internist places a stethoscope over the left side of his neck and hears a **loud bruit** over the carotid artery. Carotid duplex ultrasound shows **80% stenosis of the left internal carotid artery** with an irregular, ulcerated plaque. He has had a **transient ischemic attack (TIA)** — a "warning shot" stroke from an atherosclerotic plaque that briefly threw small emboli to his cerebral circulation.

A blood vessel wall is built in three layers; arteries and veins look similar in microscope cross-section but use the layers very differently to do very different jobs. Atherosclerosis is what happens when one of those layers slowly destroys itself over decades.

GUIDING QUESTION

What are the three layers of a blood vessel wall, how do arteries, veins, and capillaries each modify that floor plan for their job, and what is atherosclerosis doing inside Walter's carotid wall?

The shared floor plan for every vessel

Every blood vessel except the smallest capillaries has the same three-layered wall:



Tunica intima

Innermost. Single layer of *endothelium* (simple squamous) on a thin basement membrane. The smooth surface that blood flows over.



Tunica media

Middle. *Smooth muscle* and elastic fibers. Controls vessel diameter (vasoconstriction/dilation) and absorbs the pulse.



Tunica externa (adventitia)

Outermost. Connective tissue with collagen.
Anchors the vessel to surrounding tissue. Contains tiny vessels (*vasa vasorum*) that feed the larger vessel's wall.

The thickness and composition of each tunic varies enormously between vessel types, depending on what that vessel has to do.

Three sizes, three jobs

● Elastic (conducting) arteries

Aorta and its largest branches (carotids, subclavians). Thick tunica media dominated by *elastic fibers*. Stretch when the LV ejects, recoil during diastole — keeping blood flowing forward continuously. *Walter's carotid lives here.*

● Muscular (distributing) arteries

Medium-sized arteries to organs (renal, brachial, femoral). Thick tunica media dominated by *smooth muscle*. Control flow to specific tissues by changing diameter.

● Arterioles

Smallest arteries. Walls are mostly smooth muscle. Major site of resistance to flow. Their tone sets systemic vascular resistance and therefore blood pressure.

Where the work actually happens

Capillaries are the smallest vessels, just one endothelial cell thick. This is where oxygen, nutrients, hormones, and wastes cross between blood and tissue. Three types based on permeability:

● **Continuous**

Endothelial cells joined by tight junctions. Most selective. Found in muscle, skin, lung. The blood-brain barrier uses these.

● **Fenestrated**

Endothelial cells perforated by small windows (fenestrae). More permeable. Found in intestinal villi, glomerulus, endocrine glands.

● **Sinusoidal**

Wide irregular channels with large gaps and incomplete basement membrane. Most permeable. Found in liver, spleen, bone marrow. Let whole cells pass.

Thin walls, big lumens, valves to defy gravity

Veins return blood to the heart at low pressure. Compared to arteries of similar size, veins have **thinner walls, larger lumens, and less smooth muscle**. The largest veins (vena cavae) also have minimal smooth muscle.

Three mechanisms move blood through veins against gravity (in the limbs):



One-way valves

Folds of intima that prevent backflow. Failed valves = varicose veins.



Skeletal muscle pump

Contracting calf muscles compress veins and push blood upward; valves prevent backflow when muscles relax.



Respiratory pump

Inhalation lowers intrathoracic pressure, sucking venous blood toward the heart.

The decades-long disease in the vessel wall

Atherosclerosis is a chronic inflammatory disease of the **arterial intima**. It develops slowly over decades and is the underlying cause of most heart attacks, ischemic strokes, and TIAs.

Stages:

1. **Endothelial injury** from hypertension, hyperlipidemia, smoking, diabetes. The intima becomes more permeable.
2. **Lipid (LDL) accumulates** in the intima and is oxidized.
3. **Macrophages** migrate in, engulf the oxidized LDL, and become *foam cells*. They die. The lipid stays.
4. **Fatty streak** forms — a yellow lipid-laden patch in the intima.
5. **Smooth muscle cells migrate** from the media into the intima, proliferate, lay down collagen → *fibrous cap*.
6. **Atherosclerotic plaque**: fibrous cap over a necrotic lipid-rich core.
7. **Complications**: plaque rupture (exposes thrombogenic core, triggers clot), plaque hemorrhage, ulceration that throws emboli (Walter's situation), critical stenosis.

The carotid bifurcation is a hot spot. Atherosclerosis preferentially forms where blood flow is turbulent: branch points, curves, bifurcations. The carotid bifurcation is one of the most common sites in the body. The plaque grows into the lumen, narrowing it, and the ulcerated surface throws small platelet-fibrin or cholesterol emboli into the cerebral circulation, causing transient ischemic attacks.

Two systems worth knowing

PORTAL CIRCULATION

Two capillary beds connected by a portal vein. Best example: *hepatic portal system*. Blood from the GI tract first passes through capillaries in the gut wall, then is collected into the portal vein, then passes through a second capillary bed (sinusoids) in the liver before returning to the heart. This lets the liver process nutrients and detoxify substances absorbed from the gut.

ANASTOMOSES

Where two arteries (or two veins) join, providing collateral flow. Important in the brain (circle of Willis), heart (limited collaterals between coronary arteries — bad), GI tract (extensive), and joints. Anastomoses are why some arterial occlusions are catastrophic (no backup) and others are tolerated (rich collateral).



DRAW ZONE 1 -- Sketch a generic arterial cross-section. Label intima (with endothelium), media, externa. Then draw a stenotic atherosclerotic plaque next to a normal lumen.

Show the fibrous cap, necrotic lipid core, and the narrowed flow channel. Star an ulcerated surface throwing emboli.

DRAW ZONE 2 -- Diagram Walter's carotid bifurcation in the neck. Show the common carotid splitting into internal and external carotid. Mark the plaque at the bifurcation and the embolus traveling into the cerebral circulation.

The internal carotid feeds the anterior cerebral and middle cerebral arteries. Walter's right-sided symptoms = left-sided lesion (motor cortex is contralateral).

RETURNING TO WALTER

A bruit, an ulcerated plaque, and a brain that briefly lost a part

Walter's TIA is a warning. The atherosclerotic plaque in his carotid has been forming for decades; the embolus that briefly silenced his right hand and his speech is asking him to act before the next one is bigger.

"Why right-sided symptoms"

Motor and language areas in the LEFT cerebral hemisphere control the right body and dominant-hemisphere language. An embolus from the LEFT internal carotid lodges in left middle cerebral artery branches → right-sided weakness and dysarthria/aphasia.

"Why 15 minutes and fully resolved"

The embolus was small and dissolved or moved. The tissue was briefly ischemic but did not infarct. TIAs by definition resolve within 24 hours (most < 1 hour) without infarction on imaging. Walter's brain dodged a bullet.

"What is the bruit"

Turbulent flow across the stenotic plaque generates an audible whoosh. A loud carotid bruit in a patient with TIA symptoms strongly suggests carotid stenosis.

"Why ulcerated plaques are dangerous"

The ulcerated surface is exposed thrombogenic material (collagen, lipid, foam cells). It actively forms small thrombi that break off as emboli. Without ulceration, even a tight stenosis throws fewer emboli.

"Stroke risk after a TIA"

About 10-15% of patients have a stroke within 90 days; nearly half of those occur in the first two days. This is a medical urgency.

"What's the treatment"

Aggressive medical therapy (antiplatelet, statin, BP control). For 70-99% symptomatic stenosis, carotid endarterectomy (surgical plaque removal) or carotid stenting significantly reduces stroke risk. Walter is a candidate.

Intima, media, externa, plaque. Walter's carotid wall has been changing since his thirties; today is when it told him. When you go back to the prework, the artery cross-section diagram will land differently — it's the wall the surgeon will be opening.